

you are a expert statistics teacher.create a details theoretical note about PMF and CDF of PMF. seperate module.add a case study in business scenario about the interpretation and usecase also use picture visual.tone will be explanation type and soft. inter active visualization in the last

Module 1: The Probability Mass Function (PMF)

Imagine an artisan baker who prepares a single kilogram of dough every morning. She divides this entire dough ball into distinct, separate pastries: a small bun gets 100 grams, a croissant gets 200 grams, and a loaf gets 700 grams. She never spills a single crumb; every gram lands on a specific, countable item.

In statistics, a **discrete random variable** behaves just like that dough. It takes on isolated, countable outcomes, and the **Probability Mass Function (PMF)** describes exactly how the total probability "mass" of 1 (or 100%) is distributed across those outcomes.

A random variable X is discrete if its set of possible values is finite or countably infinite:

$$S_X = \{x_1, x_2, x_3, \dots\}$$

The Probability Mass Function, denoted $p_X(x)$ or $P(X = x)$, assigns a probability to each specific value x :

$$p_X(x) = P(X = x)$$

Mathematical Properties of the PMF

Every valid PMF must satisfy two non-negotiable axioms:

Non-negativity: For every possible value x , the probability cannot be negative or exceed certainty:

$$0 \leq p_X(x) \leq 1$$

Total Mass Unity: Summing the probabilities over all possible outcomes in the support must equal exactly 1:

$$\sum_{x \in S_X} p_X(x) = 1$$

If x is not an element of S_X , then $p_X(x) = 0$.

Intuitive Perspective

In modern machine learning and decision science, think of the PMF as a **lookup dictionary**. If an e-commerce algorithm needs to know the exact probability that a user clicks a button exactly 3 times, it queries $p(3)$. The PMF provides point-specific precision: it answers the exact question, *"What is the probability that X equals this exact number?"*

Module 2: The Cumulative Distribution Function (CDF)

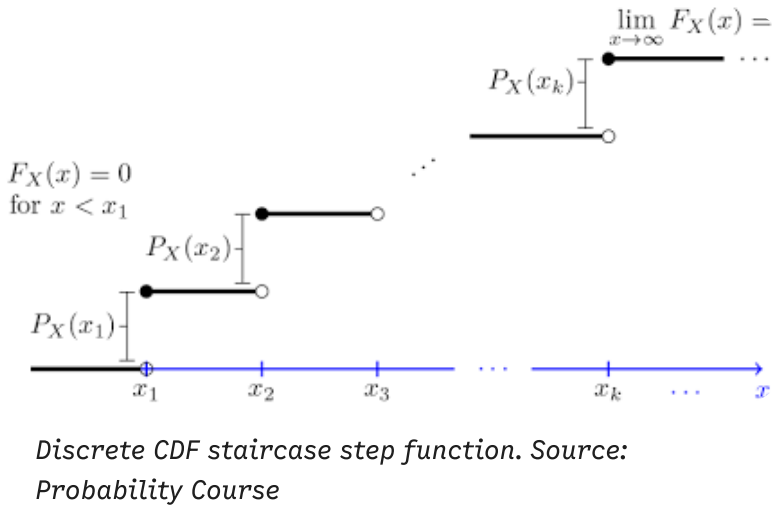
Returning to our baker, suppose a customer walks in and asks: *"How much total dough went into making items of size 200 grams or smaller?"* To answer, the baker does not just inspect

the croissant; she sweeps from left to right, gathering the bun (100g) and the croissant (200g) to give a total of 300g.

The **Cumulative Distribution Function (CDF)** performs this exact sweeping aggregation. While the PMF answers "What is the probability of $X = x$ ", the CDF answers "**What is the probability that X will be at most x ?**"

For any discrete random variable X , the CDF, denoted $F_X(x)$, is defined as:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p_X(x_i)$$



Key Structural Properties of the Discrete CDF

The diagram above illustrates the structural signature of a discrete CDF:

Staircase Step Function: $F_X(x)$ stays completely flat between isolated values because no new probability mass exists in the empty spaces between outcomes.

Jump Discontinuities: At every valid outcome x_k , the function takes an instantaneous vertical jump. The exact height of that jump equals the PMF at that point:

$$\text{Jump Height} = F_X(x_k) - \lim_{t \rightarrow x_k^-} F_X(t) = P(X = x_k)$$

Right-Continuity: Notice the solid dots on the left of each step and open circles on the right of the previous plateau. Formally:

$$\lim_{t \rightarrow x^+} F_X(t) = F_X(x)$$

This means if you land directly on the value x_k , that point's probability mass is immediately included.

Asymptotic Limits:

$$\lim_{x \rightarrow -\infty} F_X(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F_X(x) = 1$$

Interval Probability Extraction: To find the probability that X falls strictly within a range $(a, b]$:

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

Module 3: Business Case Study – Inventory Risk at CloudRoasters

Scenario: CloudRoasters supplies nitrogen cold-brew kegs on weekly contracts to local tech hubs. Demand fluctuates based on office attendance, team events, and seasonal habits. Demand X is a discrete random variable representing the number of kegs ordered in any given week, ranging from 0 to 4 kegs.

Unsold kegs spoil after one week, costing CloudRoasters \$60 per wasted keg. Meanwhile, a stockout loses \$140 in profit and damages client trust.

Weekly Keg Demand (x)	PMF: $P(X = x)$	CDF: $F_X(x) = P(X \leq x)$	Business Metric
0	0.05	0.05	5% chance of 0 demand
1	0.20	0.25	25% chance of 1 or fewer
2	0.40	0.65	65% chance of 2 or fewer
3	0.25	0.90	90% chance of 3 or fewer
4	0.10	1.00	100% chance of 4 or fewer

Decision Analysis: Interpreting PMF vs. CDF

1. Point Estimation via PMF

The company wants to know: *"Which specific batch size is most likely to be requested?"*

- Looking at the PMF, $P(X = 2) = 0.40$ is the mode. Exactly 2 kegs is the single most probable isolated event.
- However, relying solely on the PMF can be misleading. While 2 is the most likely single outcome, there is still a 60% chance ($1 - 0.40 = 0.60$) that demand will **not** be 2.

2. Service Level and Stockout Risk via CDF

The operations lead asks: *"If we brew 3 kegs, what is our service level (the probability we satisfy all customer demand without running short)?"*

- This is a cumulative question: we meet demand as long as $X \leq 3$.
- Using the CDF table directly:

$$\text{Service Level} = F_X(3) = P(X \leq 3) = 0.90 \text{ (90\%)}$$

3. Stockout Risk via Complementary CDF (Survival Function)

The risk of disappointing customers is the probability that demand strictly exceeds our inventory of 3 kegs:

$$P(X > 3) = 1 - F_X(3) = 1 - 0.90 = 0.10 \text{ (10\%)}$$

The operations manager now has an executive-level metric: stocking 3 kegs guarantees a 90% fulfillment reliability rate, with an acceptable 10% stockout exposure.

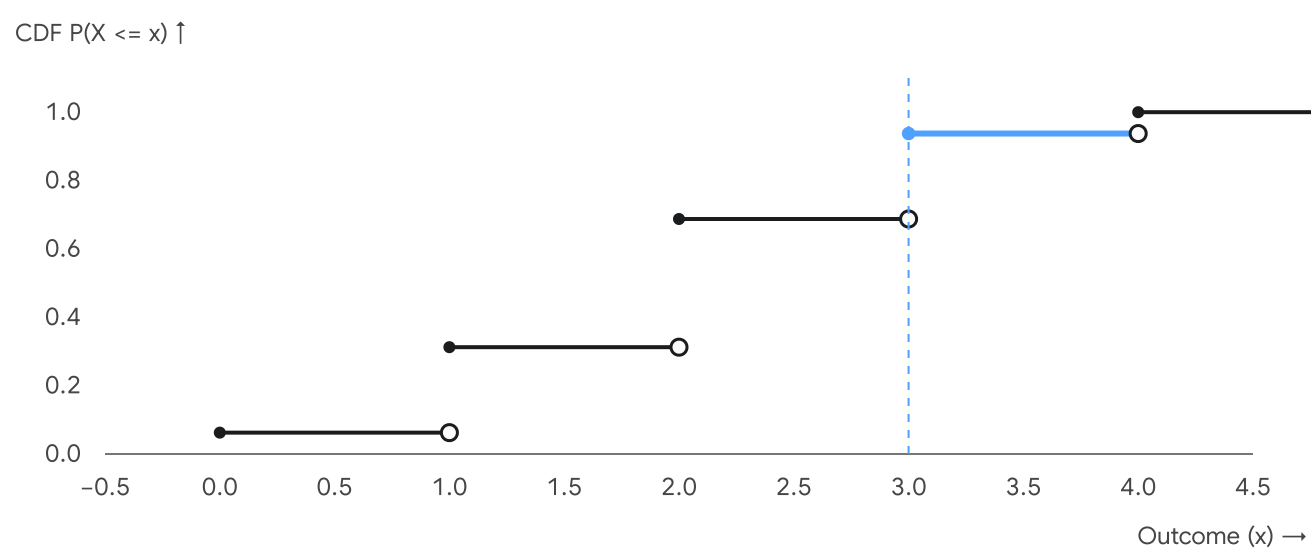
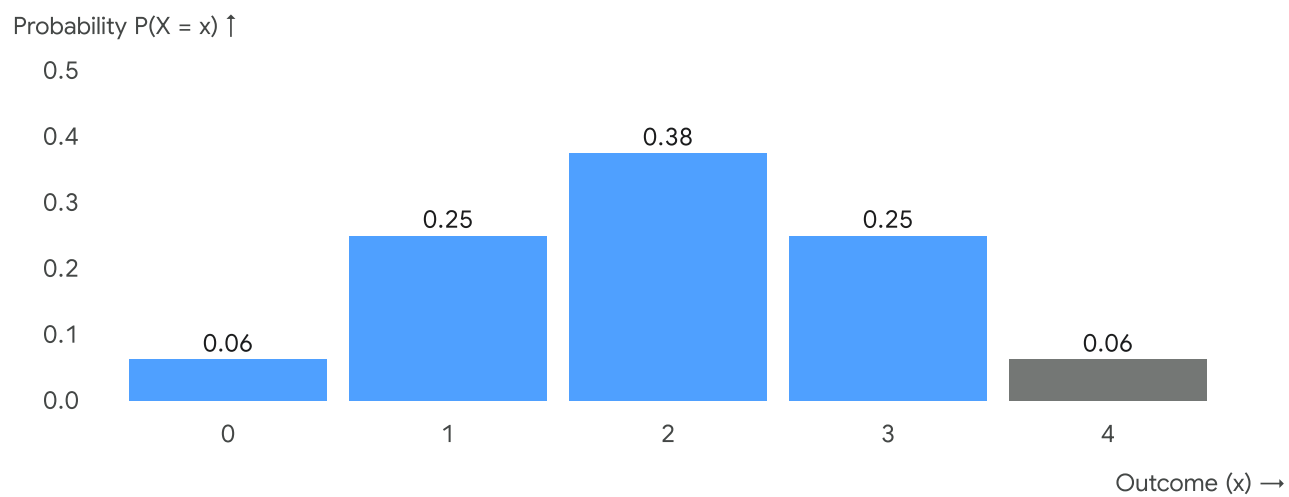
Strategic Summary for Decision Makers

- Use **PMF** when modeling specific point-in-time scenarios, diagnosing individual failure modes, or calculating expected values: $E[X] = \sum x \cdot p(x)$.
- Use **CDF** when establishing risk thresholds, setting inventory buffers, defining service level agreements (SLAs), or calculating percentiles.

Interactive PMF & CDF Explorer

Experiment with the distribution of the CloudRoasters case study below. Adjust the weekly demand threshold to inspect how individual point masses accumulate into the cumulative step function.

Discrete Distribution Explorer


$$P(X = K)$$

0.25

SERVICE LEVEL $P(X \leq K)$
0.94

STOCKOUT RISK $P(X > K)$
0.06

Threshold (k)

3

Dataset

Binomial ($n=4, p=0.5$)

Was this visual helpful?

Yes

No

Want to explore how to calculate the Expected Value and Variance from this PMF?

↪ Yes